



Everything around us is made of **Matter**!

**Matter**

It is everything that has mass and volume and occupies space.

Each matter consists of small particles called molecules and each molecules consists of Atom

**Atom**

It is the building unit of matter

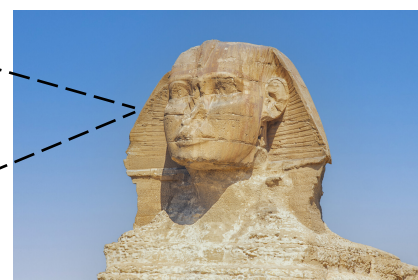
Examples :**1) The building unit of house :**

The house is formed of several floors, each floor consists of several apartments, each apartment consists of walls, each wall is formed of bricks

So the bricks is considered the building unit of the house

**2) The building unit of Sphinx :**

Sphinx is made of Limestone rocks (Matter),
which consist of Calcium carb0nate (Molecules)

**Structure of the Atom**

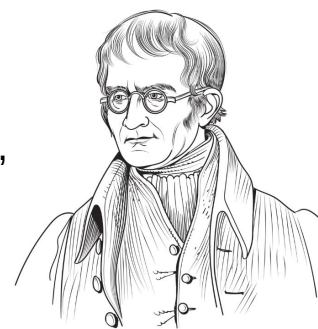
Scientists have made numerous attempts to discover the structure of the Atom

1) In ancient times

Greek philosophers believed that matter was made up of small, indivisible parts called Atoms

2) In the early 19th century

The scientist **Dalton** developed the first scientific theory about the atom,
and stated that atoms are indivisible



Dalton

3) The scientist Rutherford

He is made the first experimental model of the atom



Scientists have discovered that Atom consists of

1) Nucleus

- It is tiny space in atom located at the center of Atom
- It is very small compared to size of the Atom

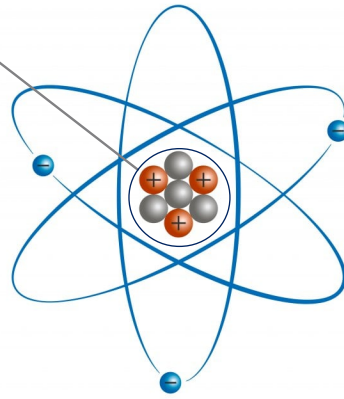
It contains two types of particles

1) Protons (+)

Positively charged particles.

2) Neutrons ($\pm$)

Neutral particles (no charge).



2) Electrons

- They are extremely small particles that revolve around the Nucleus at high speed
- They are Negatively charged ($-$)
- They are equal to protons

1) Protons, Neutrons and electrons are called subatomic particles

2) The charge of a proton is equal the charge of an electron in magnitude, but differs in type

3) If we imagine the atom as the size of a baseball field, the nucleus would be the size of a pinhead in the center of the field

Notes



Give reasons

1) The Atom is electrically neutral.

Because the number of negative electrons are equal to the number of positive protons

2) The Mass of the Atom is concentrated in the Nucleus.

Because the mass of Electron is negligible (very small) relative to Mass of nucleus

3) The nucleus has a positive charge.

Because It contains positive protons and neutral neutrons.





The properties of subatomic particles

particles	Symbol	Charge	Atomic mass unit
Proton	P^+	+1	1 u
Neutron	n^0	0	1 u
Electron	e^-	-1	$\frac{1}{1836}$ u

Chemical Symbols for elements

Scientists agreed to represent elements using chemical with symbols.

To facilitate (easy) their expression and handling, especially in chemical equations

How to write the chemical symbols ?

1 Symbols of element is represented by one letter an written in capital

➤ Such as "Fluorine"

f ✗ **F** ✓

2 If symbol was two letters, it's first letter written in Capital and second letter written in small (Na, Fe)

➤ Such as "Calcium"

CA ✗ **Ca** ✓

3 Some element have Latin name differ from their English names, so they symbolized according to their Latin letters

English	Latin	Symbol
Sodium	Natrium	Na
Potassium	Kalium	K
Copper	Cuprum	Cu
Iron	Ferrum	Fe



Examples of symbols of some famous elements

Name	Symbol	Name	Symbol	Name	Symbol
Hydrogen	H	Potassium	K	Iodine	I
Helium	He	Magnesium	Mg	Carbon	C
Mercury	Hg	Lithium	Li	Calcium	Ca
Oxygen	O	Zinc	Zn	Chromium	Cr
Fluorine	F	Nitrogen	N	Copper	Cu
Iron	Fe	Neon	N	Chlorine	Cl
Phosphorus	P	Sodium	Na	Aluminum	Al
Lead	Pb	Boron	B	Argon	Ar
Sulphur	S	Beryllium	Be	Gold	Au
Silicon	Si	Bromine	Br	Silver	Ag



The Fertilizers

Chemical compounds that are used to improve the agricultural production

- The most important types of fertilizers is : **NPK Fertilizers**
- Excessive use of fertilizers is harmful to the plant, soil, human, animals health and environment in general

Phosphorus (P)

Is essential for strengthening plant roots

Nitrogen (N)

Is necessary for the greening of plant leaves.



Potassium (K)

Is necessary for the healthy plant growth.



The Atomic number and Mass number

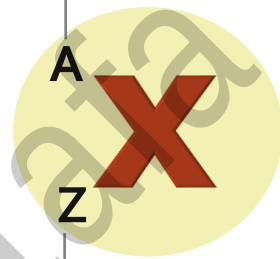
Mass Number (Nucleons)

It is the sum of number of protons and the neutrons found inside the nucleus in the element atom

Mass Number = Number of protons + Number of Neutrons

Number of Neutrons = Mass Number - Number of protons

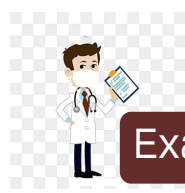
$$(n = A - Z)$$



Atomic Number

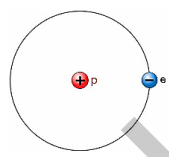
It is the sum of number of positive protons found inside the nucleus in the element atom

Atomic Number = Number of protons = Number of electrons

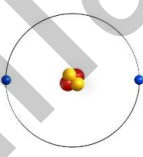


Examples

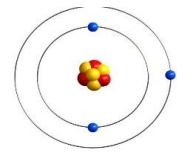
Hydrogen



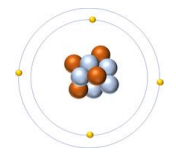
Helium



Lithium



Beryllium



	Mass number	Atomic number	Protons	Electrons	Neutrons
Hydrogen	1	1	1	1	0
Helium					
Lithium					
Beryllium					



Note

1) The number of neutrons may be equal or greater than number of protons



Electronic Configuration

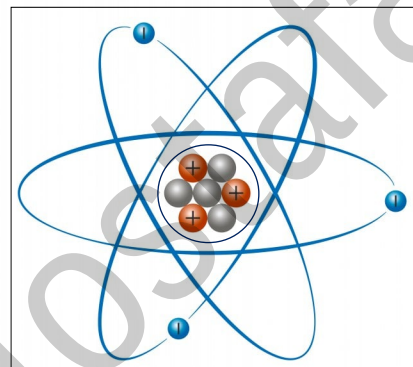
There are electron orbits in where electrons revolve around the nucleus called **energy level**



Energy levels

Imaginary regions where electrons orbit the nucleus according to their energy

- 1) The number of main energy levels is seven main levels
- 2) The energy of electron is the same energy of its level
- 3) Each main energy level consists of a specific number of **energy sub-level**
- 4) The energy level is denoted by the symbol (n) and is represented by the letters



(K, 1 L, 2 M, 3 N, 4 O, 5 P, 6 Q, 7)

Calculating the Number of Electrons in the Principal Energy level

* The number of electrons needed to fill the first four energy levels can be determined by using the formula $2n^2$ where (n) represents the number of the main energy level.

Energy level	Level number	Rule used	Number of electrons	Electrons
K	1	$2n^2$	2×1^2	2 electrons
L	2	$2n^2$	2×2^2	8 electrons
M	3	$2n^2$	2×3^2	18 electrons
N	4	$2n^2$	2×4^2	32 electrons



Notes

1) Lower energy levels are filled with electrons first, followed by higher energy levels.

For example The first level (K) is filled first, then the second level (L) and so on

2) The last (outermost) energy level in any atom doesn't contain more than 8 electrons, except (K) cannot contain more than 2 electrons only



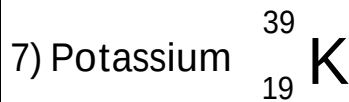
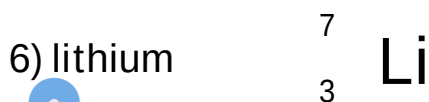
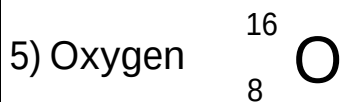
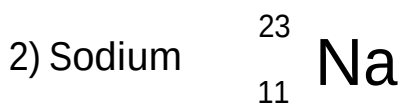
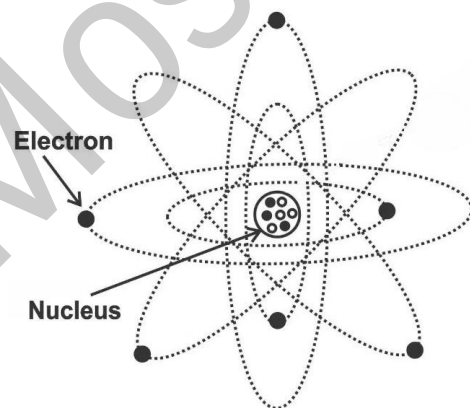
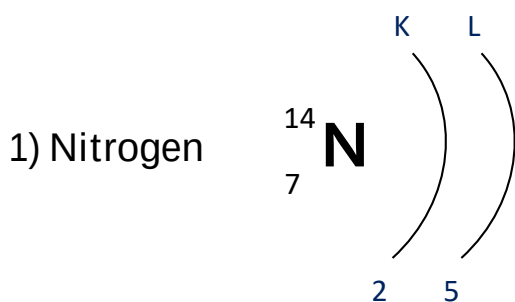
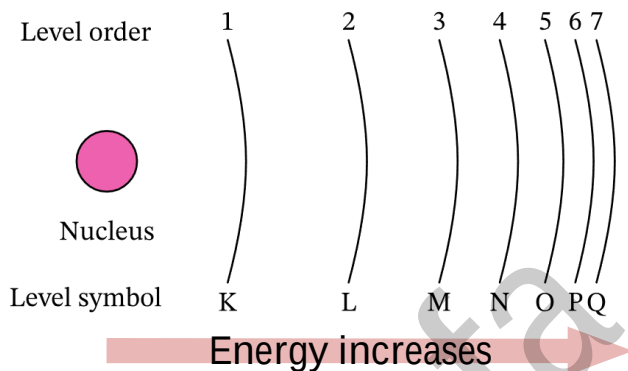
- * The energy difference between each level and the next one decreases as the distance from the nucleus increases

For example

The energy difference between level (L) and level (M) is less than

The energy difference between level (K) and level (L)

Examples : Write the electronic configuration of :-





Isotopes

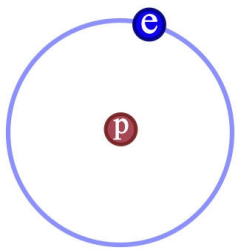


Isotopes

They are different forms of the same element have the same atomic number but differ in mass number

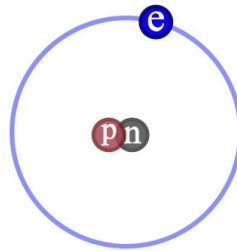
Example of Hydrogen isotopes

${}^1_1\text{H}$ Protium



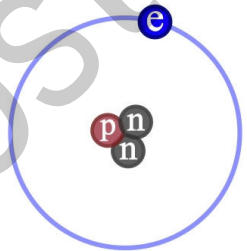
It has only one proton and no neutrons

${}^2_1\text{H}$ Deuterium



It has only one proton and one neutron

${}^3_1\text{H}$ Tritium



It has only one proton and two neutrons

- Notes**
- 1) The only isotope of hydrogen that does not have any neutrons in its nucleus is Protium
 - 2) In deuterium, the number of protons equal number of neutrons
 - 3) In tritium, the number of neutrons twice the number of protons

Example of Carbon isotopes

${}^{12}_6\text{C}$ Carbon 12

${}^{13}_6\text{C}$ Carbon 13

${}^{14}_6\text{C}$ Carbon 14

Example of Magnesium isotopes

${}^{24}_{12}\text{Mg}$ Magnesium 24

${}^{25}_{12}\text{Mg}$ Magnesium 25

${}^{26}_{12}\text{Mg}$ Magnesium 26

Give reasons

- 1) Isotopes of an element differ in their atomic mass. Due to the difference in the number of neutrons.



Give reason

Many attempts are made by scientists for classification of element.

محاولة

- To**
- Facilitate their study لكي يسهل دراستهم
 - Find relationship between chemical and physical properties of elements

The most important attempts are :-

Mendeleev's
Periodic table

Moseley's
Periodic table

Modern periodic table

1 Mendeleev's Periodic table

- He is Russian scientist
- It's the first real periodic table
- He was honored 48 years after his death by naming one of the discovered elements after him, called Mendelevium ($_{101}\text{Md}$)



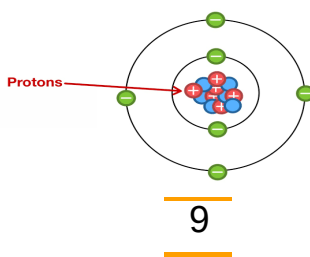
Dmitri Mendeleev

- 1) He arranged elements ascending order according to their atomic weights (mass) with irregular pattern from left to right in the horizontal rows called periods
- 2) He discovered the properties of elements were repeated periodically by the beginning of each period

2 Moseley's Periodic table

1) Rutherford :-

- He is The New-Zealand scientist
- He discovered that nucleus of atom contains positive protons





2) Moseley :-



- He is English scientist
- He Discovered that the periodicity of elements related to its atomic number not atomic weight
- He discovered that the atomic number of each element increases by one then the preceding element in the same period.
- He added zero group which includes inert gases
- He added elements which were discovered after Mendeleev.

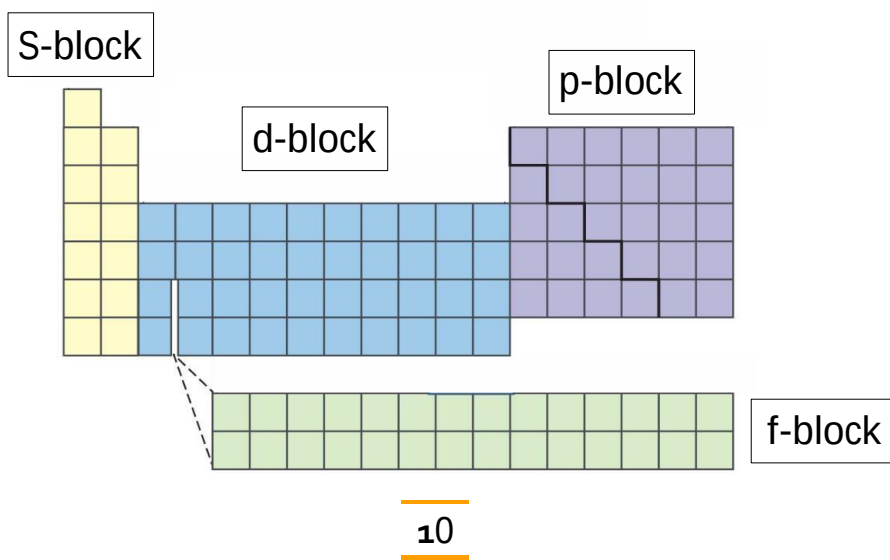
3 Modern periodic table

❖ Elements are classified in the modern periodic table in an ascending order according to :-

- ✓ Their atomic numbers.
- ✓ The way of filling the energy sublevels with electron

Description of the modern periodic table

- 1) It contains 118 elements
- 2) It consists of 7 horizontal periods (Each period starting with a new energy level)
- 3) It consists of 18 vertical groups
- 4) The element of modern periodic table are classified into 4 blocks



S-block

p-block

Zero group

gp. 1A

1	1	d-block																18
2	2																	2
3	3	f-block																10
4	4																	18
5	5																	36
6	6																	54
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Notes

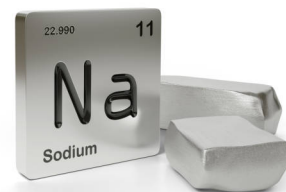
The number of Elements in each period is :

- * Period number (1) contains 2 Elements
- * Period number (2) contains 8 Elements
- * Period number (3) contains 8 Elements
- * Period number (4) contains 18 Elements

Classification of elements according to their physical state.

1) Solids :- Which are most of elements

- | | | |
|-------------------|----------------|---------------|
| 1) Lithium (Li) | 2) Sodium (Na) | 4) Iron (Fe) |
| 3) Aluminium (Al) | 5) Copper (Cu) | 6) Carbon (C) |



2) Liquids :- Which are 2 elements only

- 1) Mercury (Hg) : Only liquid metal
- 2) Bromine (Br) : Only liquid non metal



Mercury

3) gas :- Which are 11 elements classified into

6 Inert gases

- 1) Helium (He)
- 2) Neon (Ne)
- 3) Argon (Ar)
- 4) Krypton (Kr)
- 5) Xenon (Xe)
- 6) Radon (Rn)

5 Non metals gases

- 1) Hydrogen (H)
- 2) Nitrogen (N)
- 3) Oxygen (O)
- 4) Fluorine (F)
- 5) Chlorine (Cl)



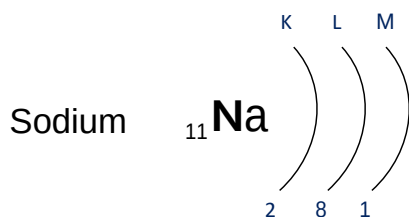
The properties of the blocks in modern periodic table

1 S-block

- 1) Located on the Left side of the periodic table
- 2) Its groups take letter (A)
- 3) It consists of two groups (1A, 2A)

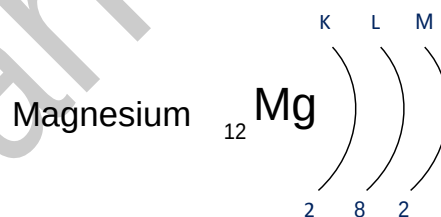
Group 1A

- 1) They are all metal except Hydrogen (Gas)
- 2) They are known as alkali metals
- 3) Their electric configuration end with 1 electron



Group 2A

- 1) They are all metal
- 2) They are known as alkaline earth metals
- 3) Their electric configuration end with 2 electrons



1A	2A
1 H Hydrogen Nonmetal	4 Be Beryllium Alkaline Earth M...
3 Li Lithium Alkali Metal	12 Mg Magnesium Alkaline Earth M...
11 Na Sodium Alkali Metal	20 Ca Calcium Alkaline Earth M...
19 K Potassium Alkali Metal	38 Sr Strontium Alkaline Earth M...
37 Rb Rubidium Alkali Metal	56 Ba Barium Alkaline Earth M...

2 d-block

- 1) Located in the middle of the periodic table between s-block (from left) and p-block (from right)
- 2) All groups take letter (B)
- 3) It consists of ten groups (3B, 4B, 5B, 6B, 7B, 8, 1B, 2B)
- 4) They begin to appear in period (4)
- 5) They are called Transition metals

3B	4B	5B	6B	7B	8			1B	2B
21 Sc Scandium Transition Metal	22 Ti Titanium Transition Metal	23 V Vanadium Transition Metal	24 Cr Chromium Transition Metal	25 Mn Manganese Transition Metal	26 Fe Iron Transition Metal	27 Co Cobalt Transition Metal	28 Ni Nickel Transition Metal	29 Cu Copper Transition Metal	30 Zn Zinc Transition Metal
39 Y Yttrium Transition Metal	40 Zr Zirconium Transition Metal	41 Nb Niobium Transition Metal	42 Mo Molybdenum Transition Metal	43 Tc Technetium Transition Metal	44 Ru Ruthenium Transition Metal	45 Rh Rhodium Transition Metal	46 Pd Palladium Transition Metal	47 Ag Silver Transition Metal	48 Cd Cadmium Transition Metal
72 Hf Hafnium Transition Metal	73 Ta Tantalum Transition Metal	74 W Tungsten Transition Metal	75 Re Rhenium Transition Metal	76 Os Osmium Transition Metal	77 Ir Iridium Transition Metal	78 Pt Platinum Transition Metal	79 Au Gold Transition Metal	80 Hg Mercury Transition Metal	

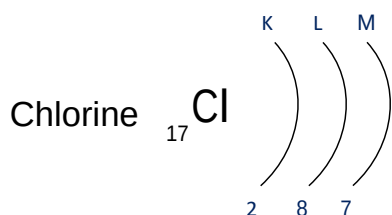


3 p-block

- 1) Located on the Right side of the periodic table
- 2) All groups take letter (A), except Zero group (Group 18 - Noble gases)
- 3) It consists of six groups (3A, 4A, 5A, 6A, 7A, Zero group)

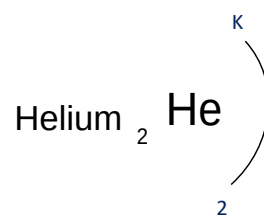
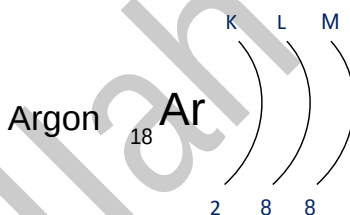
Group 7A

- 1) They are nonmetal
- 2) They are known as Halogens
- 3) Their electric configuration end with 7 electrons



Zero group

- 1) They are noble gases (inert gases)
- 2) Their electric configuration end with 8 electrons, except Helium ends with 2 electrons



Zero group

				7A	2
				F	He
				Fluorine	Helium
				Halogen	Noble Gas
3A	4A	5A	6A		
B	C	N	O		
Boron	Carbon	Nitrogen	Oxygen		
Metalloid	Nonmetal	Nonmetal	Nonmetal		
13	14	15	16	17	18
Al	Si	P	S	Cl	Ar
Aluminum	Silicon	Phosphorus	Sulfur	Chlorine	Argon
Post-Transition ...	Metalloid	Nonmetal	Nonmetal	Halogen	Noble Gas
31	32	33	34	35	36
Ga	Ge	As	Se	Br	Kr
Gallium	Germanium	Arsenic	Selenium	Bromine	Krypton
Post-Transition ...	Metalloid	Metalloid	Nonmetal	Halogen	Noble Gas
49	50	51	52	53	54
In	Sn	Sb	Te	I	Xe
Indium	Tin	Antimony	Tellurium	Iodine	Xenon
Post-Transition ...	Post-Transition ...	Metalloid	Metalloid	Halogen	Noble Gas
81	82	83	84	85	86
Tl	Pb	Bi	Po	At	Rn
Thallium	Lead	Bismuth	Polonium	Astatine	Radon
Post-Transition ...	Post-Transition ...	Post-Transition ...	Metalloid	Halogen	Noble Gas

Notes

1) p-block include :

- * All non-metals (Their electron configuration ends with 5, 6, 7 electrons)
- * All Noble gases (Their electron configuration ends with 8 electrons, except helium)
- * Some other Metals (Their electron configuration ends with 1, 2, 3 electrons)
- * All Metalloids



Metalloids

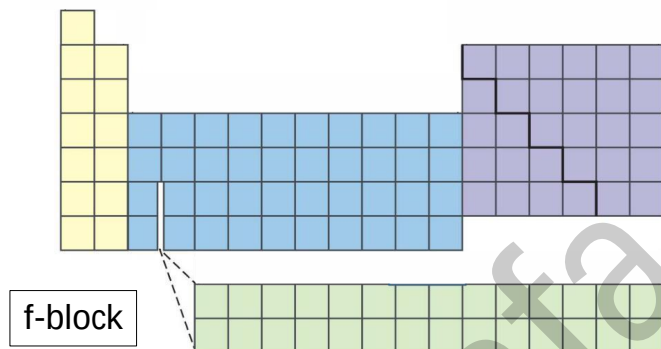
They are elements that can't be identify solely by their electric configuration due to the difference in the number of electrons in their outermost energy level

<i>Metalloids</i>	<i>Boron (B)</i>	<i>Silicon (Si)</i>	<i>Germanium (Ge)</i>	<i>Arsenic (As)</i>	<i>Antimony (Sb)</i>	<i>Tellurium (Te)</i>
<i>Period Number</i>	2	3	4	4	5	5
<i>Group Number</i>	3A	4A	4A	5A	5A	6A
<i>Electrons in outermost energy level</i>	3	4	4	5	5	6



4 f-block

- 1) They are located below the periodic table
- 2) All of them are Metals
- 3) It includes Lanthanides and Actinides



How to locate the position of an element

3	Li
Lithium Alkali Metal	
11	Na
Sodium Alkali Metal	
19	K
Potassium Alkali Metal	

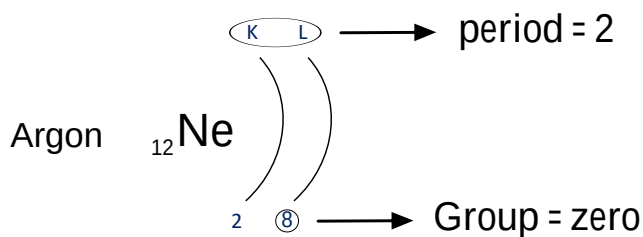
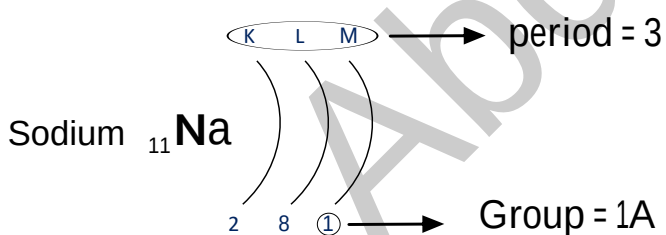
- 1) Elements (${}_3\text{Li}$, ${}_{11}\text{Na}$, ${}_{19}\text{K}$) are similar in the number of electrons in the last energy level
So they are in the same group

Period 2	6	7	8	9
	C	N	O	F
	Carbon Nonmetal	Nitrogen Nonmetal	Oxygen Nonmetal	Fluorine Halogen

- 1) Elements (${}_6\text{C}$, ${}_7\text{N}$, ${}_8\text{O}$) are similar in the number of energy levels
So they are in the same period

- 1) The number of group = number of electron in outermost energy level
- 2) The number of period = number of energy levels

For examples :-



 **Give reasons** 1) Elements of the same group share similar chemical properties.

Because They have the same number of electrons in the last energy level in each of them

- 2) Both ${}_{13}\text{Al}$ and ${}_{17}\text{Cl}$ are located in the same period in the modern periodic table.

Because They have the same number of energy level

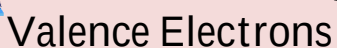


Valences may be :

- 1) Monovalent : When the number of electron in the last energy level is (1 or 7)
- 2) Divalent : When the number of electron in the last energy level is (2 or 6)
- 3) Trivalent : When the number of electron in the last energy level is (3 or 5)
- 4) Tetravalent : When the number of electron in the last energy level is (4)
- 5) Zero valence : When the number of electrons in the last energy level is (8), except helium

Lewis dot structure for valence Electrons

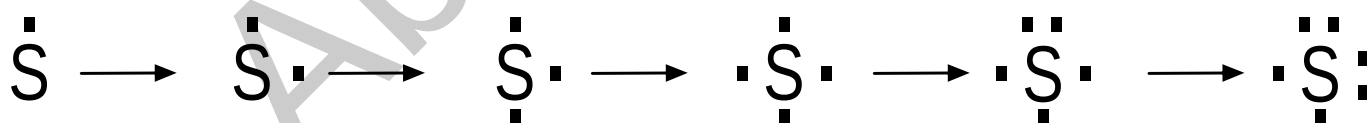
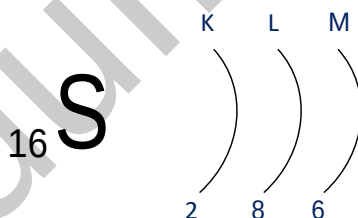
* Lewis is a scientist proposed a simplified method to represent the number of electrons which are found in the last energy level around the element symbol, known as Valence Electrons



They are electrons found in the last energy level of the element atom

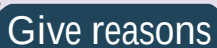
* The outer level electrons are distributed individually first on the four sides of the element symbol then pairing begins until they are distributed

For example :- Sulphur



Notes

- 1) The valence of an element = the number of the unpaired electrons in its Lewis structure



- 1) The valence of Nobel gases equal zero.

Because The last energy level is completely filled with electrons and they don't have unpaired electrons



Electron configuration of Atoms reveals their properties

* Chemical properties of element

Changes due to the difference in the number of electrons in the outermost energy level

* Physical properties of element

Changes due to the difference in the number of neutrons in the nucleus of the atom

Alkali group metal (Group 1A)	
1) Atomic radius	* It is measured by Picometer (pm) * Atomic radius <u>Increases</u> by increasing atomic number (from top to bottom) Examples :- Lithium (Li) , Sodium (Na) , Potassium (K) 157 Pm 191 Pm 235 Pm
2) Melting point and Boiling point	* Melting point and boiling point <u>decreases</u> by increasing atomic number Examples :- Lithium (Li) , Sodium (Na) , Potassium (K)
3) Chemical activity	* Chemical activity <u>Increases</u> by increasing atomic number (from top to bottom) * Cesium is the most active metal

1A

3

Li

Lithium
182

11

Na

Sodium
227

19

K

Potassium
275

37

Rb

Rubidium
303

55

Cs

Cesium
343

Halogen group (Group 7A)	
1) Atomic radius	* It is measured by Picometer (pm) * Atomic radius <u>increases</u> by increasing atomic number (from top to bottom) Examples :- Chlorine (Cl) , Bromine (Br) , Iodine (I) 99 Pm 114 Pm 133 Pm
2) Melting point and Boiling point	* Melting point and boiling point <u>increases</u> by increasing atomic number Examples :- Chlorine (Cl) , Bromine (Br) , Iodine (I)
3) Chemical activity	* Chemical activity <u>decreases</u> by increasing atomic number (from top to bottom) * Fluorine is the most active nonmetal

7A

9

F

Fluorine
135

17

Cl

Chlorine
175

35

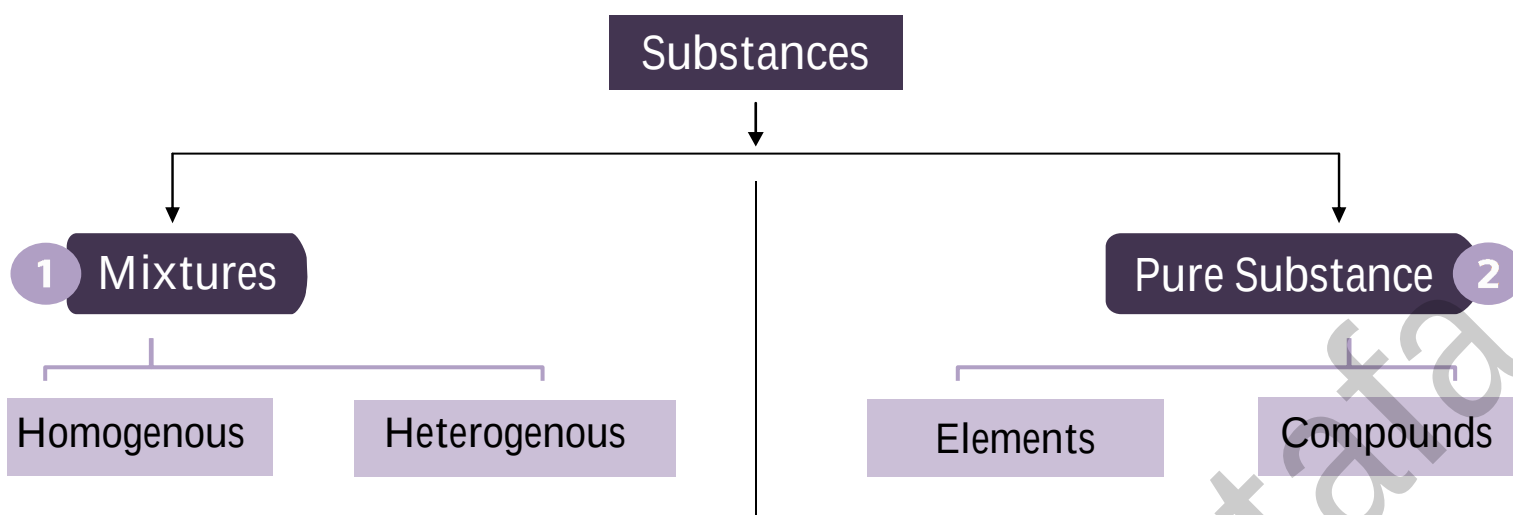
Br

Bromine
183

53

I

Iodine
198



1 Mixtures

They are substances composed of two or more substances that are not chemically combined and their components can be separated by physical methods

Physical separation methods

1 Magnetic separation

2 Filtration

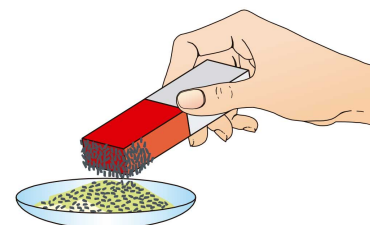
3 Evaporation and condensation

1 Magnetic separation

It is a method used to separate between magnetic material and non magnetic material

For examples :-

- * Iron filings and Sulphur
- * Iron filings and Sand



2 Filtration

It is a method used to separate a solid substance which is not dissolved in water by using a filter paper in a filtration funnel

For example :-

Heterogeneous Mixtures (Sand in Water)

Heterogeneous Mixtures

It is a mixtures in which the components can be distinguished by the naked eye





3 Evaporation and condensation

It is a method used to separate the components of a solution of a solid substance which is dissolved in water

For example :- Homogeneous Mixtures
(Salt in Water)

Homogeneous Mixtures

It is a mixtures in which the components cannot be distinguished by the naked eye



2 Pure Substance

They are Substances whose components cannot be separated by the physical methods

Pure substance may be

Elements

It is the simplest pure form of matter and cannot be dissociated into simpler forms, either by physical or chemical methods

For examples :-

* Oxygen (O_2)

* Mercury (Hg)

Molecules of elements :-

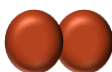
1) Mono-atomic : Consists of one atom only

Such as :- Carbon (C)



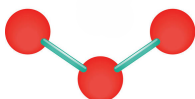
2) Diatomic : Consists of two atoms

Such as :- Oxygen (O_2)



3) Poly-atomic : Consists of many atoms

Such as :- Ozone (O_3)



Compounds

It is a pure substance formed by The combination of two or more elements in a fixed mass ratio

For examples :-

* Red Mercury Oxide (HgO)

* Water (H_2O)



Note

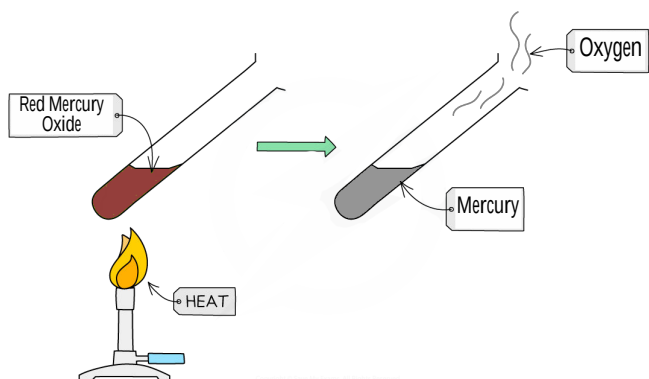
1) Their components can be separated by chemical methods

Such as :-
1) Heating
2) Electrolysis



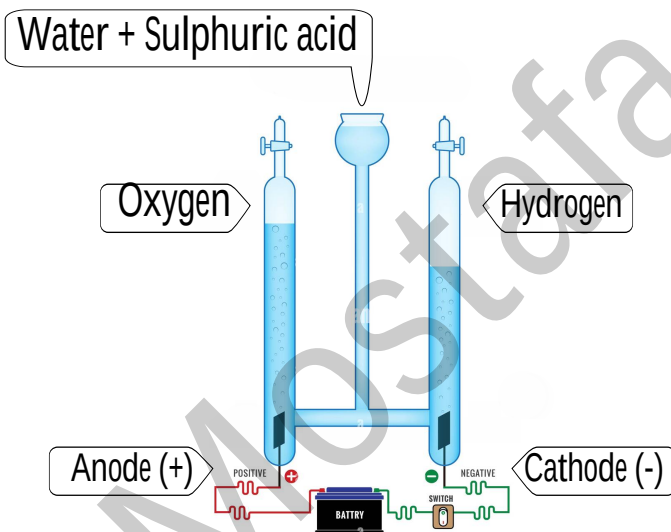
1) Heating

- * When red Mercury Oxide is heated, it decomposes into its elements (Mercury and Oxygen gas)



2) Electrolysis (The Hoffman voltammeter)

- * It is used to decompose acidified water into its elements (Hydrogen and Oxygen)



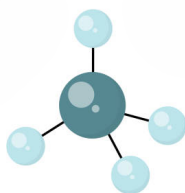
Molecules of Compounds

Organic Molecules

They are compounds in which carbon atoms are bonded mainly to hydrogen atoms or may be bonded to other atoms such as Oxygen and Nitrogen

For examples :-

1) Methane molecule (CH_4)

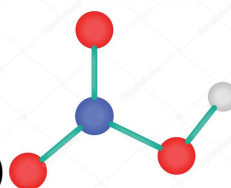


Inorganic Molecules

They are compounds which contain various elements including carbon in some cases

For examples :-

1) Nitric acid molecule (HNO_3)



2) Carbon dioxide (CO_2)



1) Organic Molecules known as Carbon compounds

2) The number of atoms in a single molecule can reach several thousands, Such as :

- * Plastic polymers
- * Hemoglobin in blood
- * Vitamin (D) : which regulates Calcium and Phosphorus level in blood to prevent Osteoporosis

Notes



Molecular formula

It is a symbolic formula that expresses the type and number of atoms of the elements that form the molecule

How to calculate the number of elements and atoms

Count the capital letters = Elements

Count the lower numbers = Atoms

Name	Ammonia	Methane molecule	Sulphuric acid	Sodium Nitrite	Nitric acid
Molecular Formula	NH_3	CH_4	H_2SO_4	NaNO_2	HNO_3
Number of Elements	2	2	3	3	3
Number of Atoms	$1 + 3 = 4$	$1 + 4 = 5$	$2 + 1 + 4 = 7$	$1 + 1 + 2 = 4$	$1 + 1 + 3 = 5$

Life Applications :- (Egyptian Blue Dye)

- ➡ It's a chemical formula is $\text{CaCuSi}_4\text{O}_{10}$
- ➡ The ancient Egyptians used it to color Papyri and Statues
- ➡ It is still used to color the house facades in Nubian villages



1) Number of Elements of Egyptian Blue Dye

Calculate ?

.....

2) Number of Atoms of Egyptian Blue Dye

.....

Properties of Substances

1 Physical properties

2 Chemical properties

Physical properties

They are Properties of the substance that can be observed and measured in some cases

- Such as :-
- 1) Viscosity
 - 2) Density
 - 3) Melting point
 - 4) Color, taste, odour
 - 5) Solubility in water
 - 6) Boiling point
 - 7) Hardness
 - 8) Electrical conductivity
 - 9) Thermal conductivity

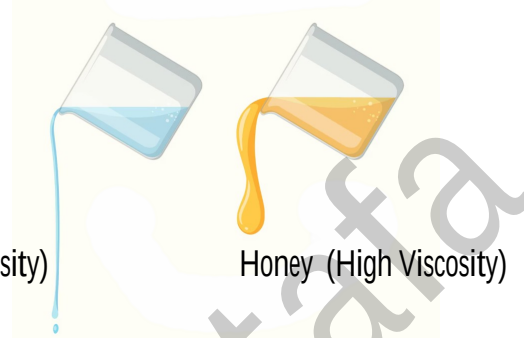


1) Viscosity

It is a physical property of liquid that describes their resistance to flow and the movement of objects through them

- ➡ The viscosity of water is less than the viscosity of honey,
So Water easier to stir than Honey

Water (Low Viscosity)

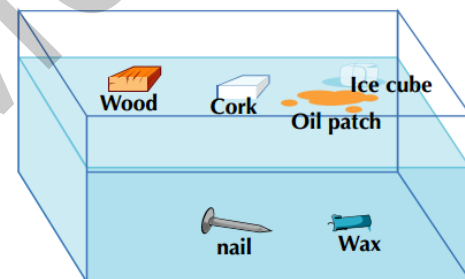


Honey (High Viscosity)

2) Density

It is a physical property used to distinguish between materials that float on water or sink in water

- ➡ The materials that have density less than density of water
(Float on its surface)
- ➡ The materials that have density greater than density of water
(Sink in water)



Give reasons

- 1) Cork floats on the water surface. Because it has density less than density of water
- 2) Iron sink in the water. Because it has density greater than density of water

3) Melting point

It is the temperature at which the state of a substance starts to change from solid state to liquid state

- ➡ The butter block melts faster than aerogel sheet



Give reasons

- 1) Aerogel sheet has high melting point.

Because it can withstand very high temperatures





Aerogel

It is a transparent, low density material, whose 99.8% of its composition is of air only



- It is the lightest solid material known to date
- It is highly durable and has excellent insulating properties
- It is used to make jackets for researchers in Antarctica



Give reasons

1) Aerogel is used to make jackets for researchers in Antarctica.

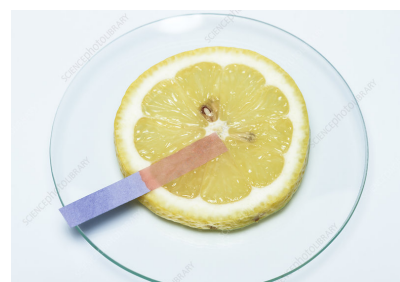
Because it works as substitute for the fur of polar bears to protect them from extinction

Chemical properties

They are Properties of the substance that only appear when a chemical reaction occurs, leading to a change in the shape and the composition of the substance

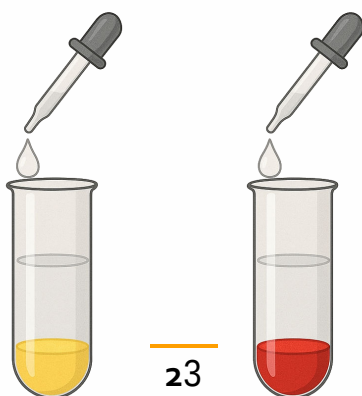
1) Effect of litmus paper

- Some substances change litmus paper into **Red**
Such as :- Lemon
- Some substances change litmus paper into **Blue**
Such as :- Toothpaste
- Some substances don't affect the color of litmus paper



2) Forming precipitate

- It is possible to differentiate between two colorless solutions, so by adding a reagent to each of them, it forms a precipitate with different colors with each of them





Uses of Substances Based on Their Properties

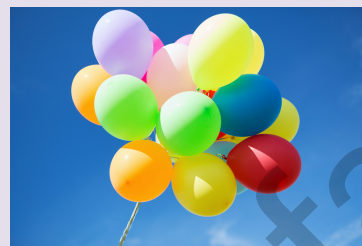
Helium (He)
(Nobel gas)

➡ It is used to fill balloons

Give reason

1) Helium is used to fill balloons.

Because Its density is less than air and It is non-flammable material



Nitrogen (N)
(Non-metallic
gas)

➡ It is used to fill car tires

Give reason

1) Nitrogen is used to fill car tires.

Because It is not affected by temperature changes and
does not react with rubber



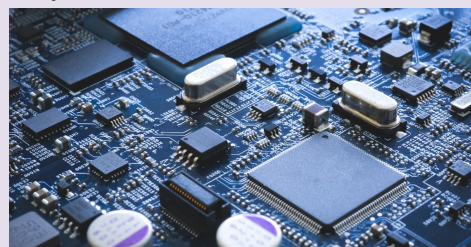
Silicon (Si)
(Metalloid)

➡ It is used to manufacture of computer chips

Give reason

1) Silicon is used to manufacture of
computer chips.

Because It conducts electricity less than metals
and more than nonmetals



Stainless steel
alloy

➡ It is used to manufacture of cooking utensils

Give reason

1) Stainless steel alloy is used to manufacture
of cooking utensils.

Because It has resistance to rusting



Aluminum-
Titanium alloy

➡ It is used in construction of military aircraft frames (Structures)

Give reason

1) Aluminum-Titanium alloy is used in
construction of military aircraft frames (Structures).

Because It is lighter than aluminum alone and retains its strength
at high temperatures





Remember what we studied previously that

1 Metals	2 Non-metals	3 Noble gases
The last energy level ends with 1,2 or 3 electrons	The last energy level ends with 5, 6 or 7 electrons	The last energy level is saturated with electrons
<u>During chemical reactions</u>		
- Metals tend to lose their valence electrons To reach the stable electron configuration - Metals form Positive Ion (Cation)	- Non-metals tend to gain one or more electrons To reach the stable electron configuration - Non-metals form Negative Ion (Anion)	- Noble gases don't lose or gain during chemical reaction - Noble gases don't form Cation or Anion

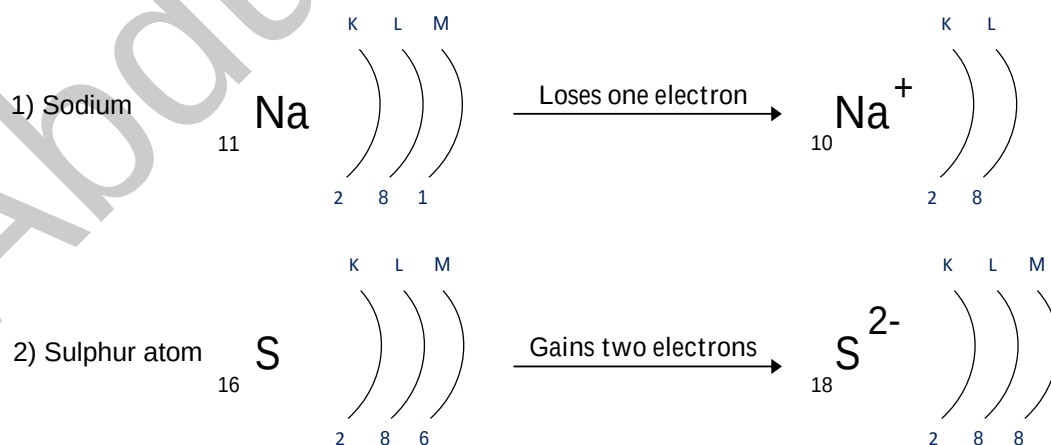
Positive Ion (Cation)

It is Metals atom that has lost one or more electrons

Negative Ion (Anion)

It is Non-metals atom that has gained one or more electrons

For examples :-



1) Noble gases do not bond under normal conditions with each other or with other elements.

2) Noble gases are stable elements.

Because outermost energy level is complete with 8 electrons, except helium, which complete with 2 electrons.

3) When an atom changes into an ion the number of nucleons don't change, while only the number of electrons changes.

Notes



Chemical bonds

Ionic bond

Covalent bond

1) Ionic bond

It is electrostatic attraction between positive ion (Cation) and negative ion (Anion) forming an ionic compound molecule

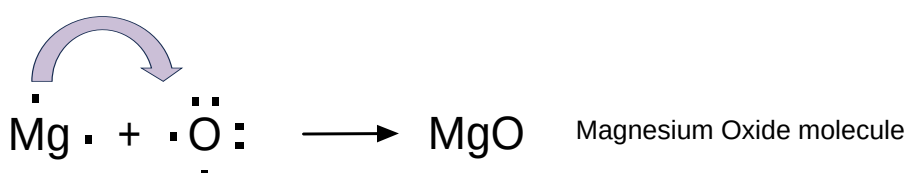
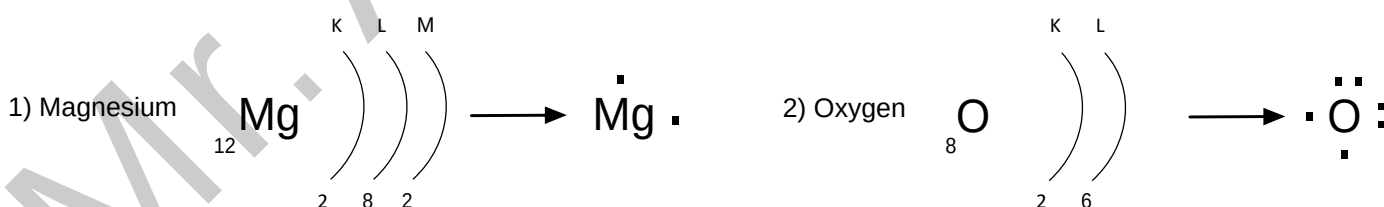
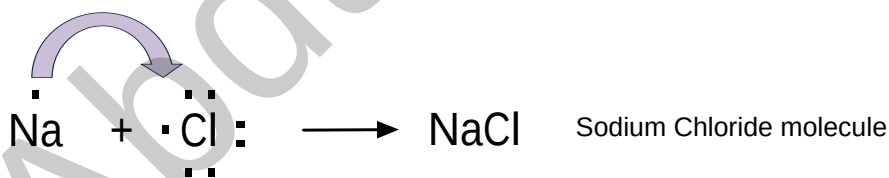
How Ionic bonding occurs ?

❖ When most metals react with nonmetals,

Metal atom loses its valence electrons forming (Cation) and Nonmetals gain electrons forming (Anion)

➡ Then they attract together due to their difference in electric charge forming ionic compound molecule

For examples :-





Give reasons

1) Ionic compound is neutrally charged.

Because the number of positive charges equals the number of negative charges in it.

2) Ionic bond produces compounds molecules only not produce elements molecules.

Because it occurs between two different atoms (Cation of metals and Anion of nonmetals)

3) It is impossible to combine sodium and magnesium together to form a compound.

Because each of them is a metal and they tend to lose their valence electrons during chemical reactions

properties of ionic compounds

1

Most of them dissolve in water

2

They conduct electricity for both their aqueous solutions or in their molten state

3

Their melting and boiling points are high

2) Covalent bond

It occurs between two atoms of the same nonmetal element or between two atoms of different nonmetal elements by sharing electrons

How Covalent bonding occurs ?

- ❖ When each atom shares a number of electrons equal to the number of electrons needed to complete its outermost energy level, without losing or gaining electrons

Types of covalent Bonds

1

Single Covalent bond

(—)

2

Double Covalent bond

(=)

3

Triple Covalent bond

(≡)

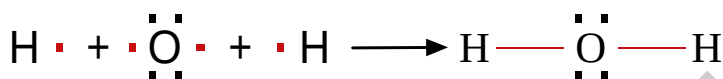


1 Single Covalent bond (—)

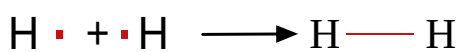
- ❖ Each atom shares its valence electron to form a pair of electrons which forms (One Single Covalent Bond)

Such as

1) Water Molecule :-
(H — O — H)



2) Hydrogen Molecule :-
(H — H)



2 Double Covalent bond (≡)

- ❖ Each atom shares its valence electron to form a pair of electrons which forms (One Double Covalent Bond)

Such as

1) Oxygen Molecule :-
(O = O)

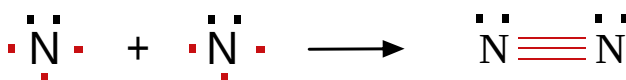


3 Triple Covalent bond (≡)

- ❖ Each atom shares its valence electron to form a pair of electrons which forms (One Triple Covalent Bond)

Such as

1) Nitrogen Molecule :-
(N ≡ N)



How does the connection between ?

1) Sodium chloride

.....

2) Hydrogen chloride

.....



Give reasons

1) Covalent bond may result in the formation of element molecule.

Because it can occurs between two atoms of the same nonmetal elements

2) Covalent bond may result in the formation of compound molecule.

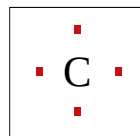
Because it can occurs between two different nonmetal atoms

properties of Covalent compounds

- Most of them don't dissolve in water
- They mostly don't conduct electricity
- Their melting and boiling points are low

Unique Properties of Carbon Atoms

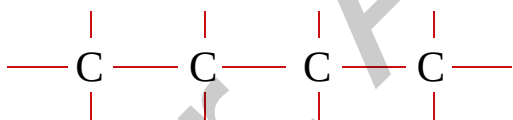
- ❖ The last energy level in carbon atom contains 4 single (Unpaired) electrons
- ❖ Carbon atoms have very unique properties



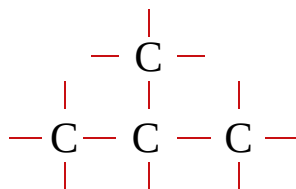
Because of their ability to bond with each other in organic compounds

Carbon bond in one of three forms

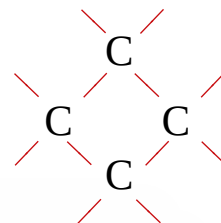
1) Straight (continuous) chains



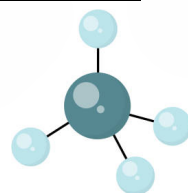
2) Branched chains



3) Cyclic structure (Cycles or Rings)



Methane molecule (The simplest organic compound)
(CH₄)



- ❖ It consists of one carbon atom is bonded to four hydrogen atoms through four single covalent bonds

